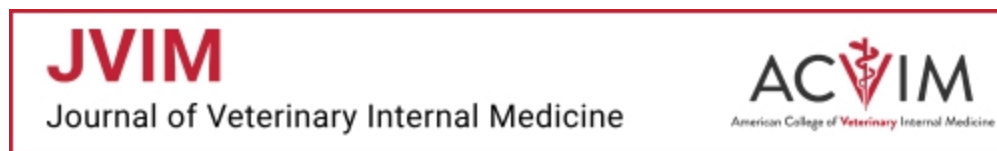


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Co-occurrence of nephrocalcinosis and nephrolithiasis in feline kidneys

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Abstract

Background

Nephrocalcinosis is common in cats with nephrolithiasis. Understanding the association between these disorders might facilitate understanding of their pathogenesis.

Hypothesis/Objectives

Nephrocalcinosis is associated with nephrolithiasis.

Animals

Kidneys were evaluated postmortem in 175 client-owned cats.

Methods

An observational cross-sectional study. High-resolution microradiography identified nephroliths and classified parenchymal mineralization as striated or punctate. Percentage mineralization was quantified on microradiographs. Associations of nephrocalcinosis and nephrolithiasis at cat level (disease in either the left or right kidney) and individual kidney level were estimated using chi-squared tests and logistic regression with odds ratios (ORs) and 95% confidence intervals (CIs).

Results

Nephrolithiasis was present in 54% of cats (95/175) and nephrocalcinosis in 84% (147/175). Cats with nephroliths were older than those without (median, 15 vs 13 years; $P = .004$). At the cat level, nephrocalcinosis was associated with higher odds of nephrolithiasis (OR, 3.5; 95% CI, 1.6-7.4; $P = .002$), but the association was attenuated after age adjustment (OR, 3.1; 95% CI, 1.0-10.1; $P = .05$; $n = 115$). At the kidney level, striated mineralization was associated with nephrolithiasis (OR, 13.9; 95% CI, 2.0-95.9; $P = .01$), whereas punctate mineralization was not significantly associated with nephrolithiasis. Percentage mineralization was higher in cats with nephroliths than in those without it ($P = .004$).

Conclusions and clinical importance

Nephrocalcinosis, particularly striated medullopapillary patterns, was associated with nephrolithiasis in cats. Pattern-specific mineralization may represent a preventive target for the management of nephrolithiasis.

Keywords: calcium oxalate, calcium phosphate, mineralization, radiography, uroliths

Introduction

Nephrocalcinosis and nephrolithiasis are distinct forms of mineralization affecting feline kidneys.^{[1](#)} Nephrocalcinosis is the deposition of calcium salts within the renal parenchyma, whereas nephrolithiasis involves the formation of discrete mineral concretions within the renal pelvis. In aged cats with chronic kidney disease (CKD), nephrocalcinosis is present in 61%-78% of cases.^{[2,3](#)} The prevalence of nephrolithiasis in cats with CKD is somewhat lower, up to 51%.^{[4](#)} Both nephrocalcinosis and nephrolithiasis share risk factors such as hypercalcemia and hypercalciuria,^{[3-7](#)} but the relationship between nephrocalcinosis and nephrolithiasis in cats remains incompletely understood.

In humans, one proposed mechanism promoting nephrolith formation involves a specific form of nephrocalcinosis, calcium phosphate (CaP) deposits that develop within the basement membrane of the loop of Henle or the collecting ducts.^{8,9} When these deposits extend toward the papillary tip and become exposed to urine in the renal pelvis, they serve as sites for heterogeneous nucleation.¹⁰ By providing a surface for initial crystal formation, these deposits facilitate nephrolith formation at lower levels of urine supersaturation.¹¹ Whether similar mechanisms contribute to nephrolithiasis in cats is unknown.

Our retrospective study evaluated the co-occurrence of nephrocalcinosis and nephrolithiasis in cat kidneys. We also assessed whether specific patterns of nephrocalcinosis are associated with nephrolithiasis. We hypothesized that nephrolithiasis occurs more commonly in cats with nephrocalcinosis.

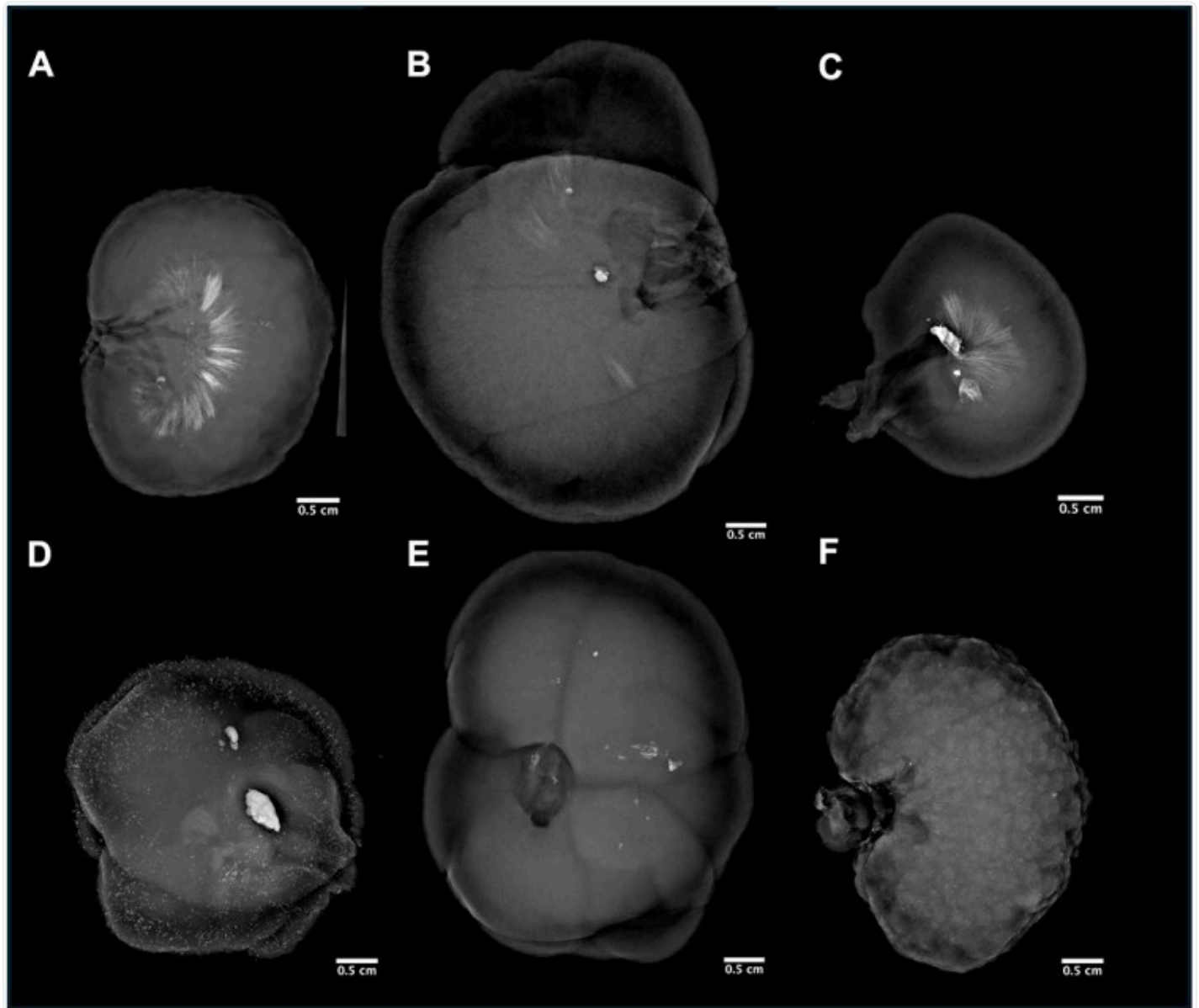
Materials and methods

An observational cross-sectional study estimated the co-occurrence of nephrolithiasis and nephrocalcinosis in cats using kidneys collected postmortem between July 1, 2016 and December 31, 2024 at a veterinary teaching hospital ([Supplementary Appendix A](#)). Donors included client-owned cats from both referral and primary care services. Kidneys were harvested immediately after euthanasia or within 24 h from refrigerated carcasses (4°C) and preserved in 10% formalin for ≥ 24 h until imaged and processed. The project was exempt from Institutional Animal Care and Use Committee oversight because of the use of cadaveric tissues. The study cohort included cats with both kidneys available. Kidneys with severe autolysis or gross distortion were excluded. Demographic data were recorded when available.

Microradiography

Microradiography was performed on whole kidneys or the remaining portions if a 2- to 3-mm central section had been removed for a separate study. Nephrolithiasis was defined as a clearly demarcated radiopaque structure within the renal pelvis or diverticula and was classified based on its appearance as smooth or irregular ([Figure 1](#)). Nephrocalcinosis was defined as unstructured parenchymal radiopacities and was classified as punctate or striated, with distribution recorded as diffuse or regional ([Figure 1](#)). Percentage mineralization was quantified on microradiographs of whole kidneys (ImageJ2 v2.14.0/1.54f) after excluding nephrolith regions ([Supplementary Appendix A](#)):

Figure 1.



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Microradiographs of feline kidneys displaying distinct mineralization patterns and nephroliths. (A) Diffuse striation and faint diffuse punctate with a smooth stone. (B) Regional striation with a smooth stone. (C) Diffuse striation and faint diffuse punctate with irregular stones. (D) Regional striation and diffuse punctate with smooth stones. (E) Regional punctate with an irregular stone. (F) Diffuse punctate with faint diffuse striation, without a stone.

$$\% \text{ Mineralization} = \frac{\text{Integrated density of mineralization}}{\text{Integrated density of whole kidney}} \times 100$$

For cats with 2 kidneys suitable for quantification, percentage mineralization was averaged to a single cat-level value. For cats with only one kidney suitable for quantification, that value was used to represent the cat level value.

To verify percentage mineralization, a random subset of kidneys was evaluated using histopathology (von Kossa-stained section; $n = 32$) or microcomputed tomography (micro-CT; $n = 13$; [Supplementary Appendix A](#)). The composition of nephroliths from 10 cats was analyzed using optical crystallography and Fourier transform infrared spectroscopy or energy dispersive spectroscopy (EDS).

Statistical analysis

Analyses were performed using R (v4.3.2). Data were summarized as counts and percentages, mean $\pm$ SD, or median (IQR). Group comparisons used chi-squared or Fisher's exact tests, t -tests, and analysis of variance (ANOVA) with Tukey's test or Wilcoxon/Kruskal–Wallis with Dunn's test, as appropriate. Kidney level logistic regressions were used to estimate associations of nephrolithiasis with sex, age, and nephrocalcinosis. Mixed effects models, incorporating a cat level random intercept, were used to account for clustering of the 2 kidneys per individual. Validation was assessed using Kendall's τ . Results are reported as odds ratios (ORs) with 95% CI; $P < .05$ was considered statistically significant ([Supplementary Appendix B](#)).

Results

Microradiographs of both kidneys were obtained from 175 cats (350 kidneys). Age and sex data were available for 115 cats. The median age was 14 years (range, 1-20 years; IQR, 10-16.5). Of these, 54% (62/115) were female (7 intact, 55 spayed) and 46.1% (53/115) were male (5 intact, 48 neutered). Breed was available for 109 cats: domestic shorthair ($n = 72$), domestic longhair ($n = 14$), domestic medium hair ($n = 14$), Siamese ($n = 4$), and one each of Birman, Ragdoll, Russian Blue, Snowshoe, and Tonkinese.

Prevalence of nephrolithiasis in cats

Nephroliths were identified in 54% (95/175) of cats ([Table 1](#)). The median age for cats with nephroliths was 15 years (range, 2-19 years; IQR, 12-17; $n = 66$). The median age for cats without nephroliths was younger at 13 years (range, 1-20 years; IQR, 6-15; $n = 49$; $P = .004$). Sex did not differ between those with and without nephrolithiasis ($P = .13$).

Table 1.

Characteristics of cats without and with nephroliths.

Variables		No nephroliths ($n = 80$)	n	Nephroliths ($n = 95$)	n	P - value
Age (median [IQR])		13 [6, 15]	49	15 [12, 17]	66	.004
Gender (% [n])	Female	45% (23)	51	61% (39)	64	.13
	Male	55% (28)		39% (25)		
Breed (% [n])	Dlh	11% (5)	45	14% (9)	64	
	Dmh	9% (4)		16% (10)		
	Dsh	73% (33)		61% (39)		
	Birman	2% (1)				
	Ragdoll			2% (1)		
	Russian Blue	2% (1)				
	Siamese	2% (1)		5% (3)		
	Snowshoe			2% (1)		
	Tonkinese			2% (1)		

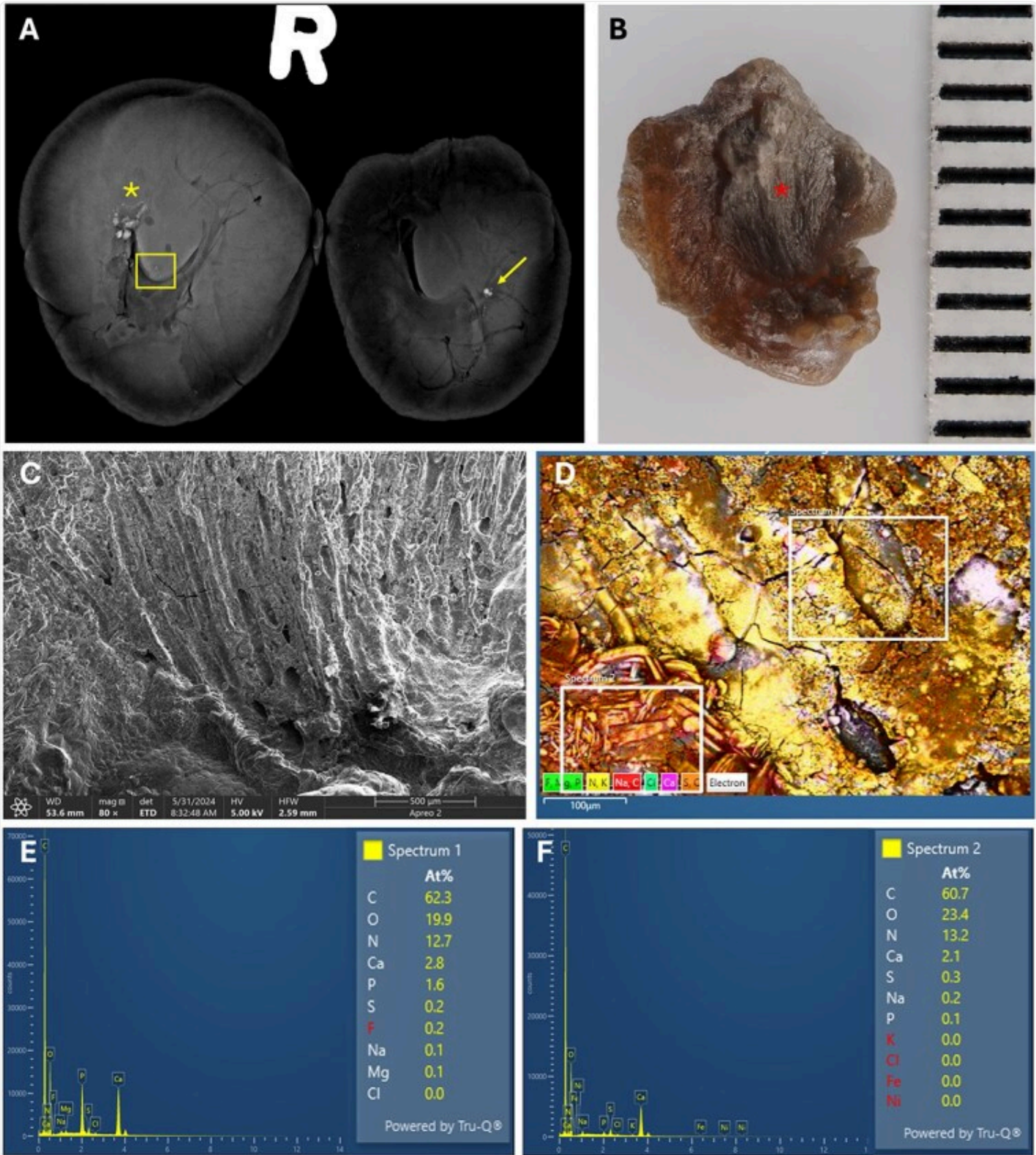
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Abbreviations: Dlh = domestic longhair; Dmh = domestic mediumhair, Dsh = domestic shorthair.

Among the cats with nephroliths, 57% (54/95) had bilateral nephrolithiasis; 76% (41/54) of these had concordant nephrolith appearance in both kidneys (33 irregular, 6 smooth, 2 mixed). Discordant bilateral patterns included one kidney with irregular nephroliths and the contralateral kidney with smooth nephroliths ($n = 10$), and 1 kidney with irregular nephroliths and the contralateral kidney with a mixed appearance ($n = 3$). Unilateral nephrolithiasis occurred in 43% (41/95) of cats ([Table S1](#)).

Among the 10 cats with nephrolith composition analysis, smooth nephroliths were composed of calcium oxalate (CaOx) monohydrate ($n = 4$), and irregular nephroliths were composed of calcium phosphate (CaP; $n = 5$). One cat had smooth and irregular nephroliths within the same kidney; composition analysis identified a smooth CaOx nephrolith and an irregular CaP nidus with a CaOx body ([Figure 2](#), [Table S2](#)).

Figure 2.



(A) Microradiograph of a feline kidney; a 2- to 3-mm central section was removed prior to imaging. Remaining portions show smooth (arrow) and irregular (asterisk) nephroliths. The boxed region indicates the site from which a nephrolith was removed for analysis. Striated mineralization is visible within this region. (B) A tubular structure on the concave surface of the stone (asterisk) was identified in (B) and confirmed by scanning electron microscopy (C). (D) EDS on carbon-coated nephroliths demonstrated that (E) Spectrum 1, obtained from the tubular structure, corresponded to a calcium phosphate-rich region. (F) Spectrum 2, obtained from the plate-like crystalline structure, contains calcium with negligible phosphorus, consistent with calcium oxalate. The weight percentage is reported in [Table S2](#). Abbreviation: EDS = energy-dispersive spectroscopy.

Prevalence of nephrocalcinosis in cats

Nephrocalcinosis was identified in 84% (147/175) of cats. The median age of cats with nephrocalcinosis was 15 years (range, 1-20 years; IQR, 12-17; $n = 95$). The median age of cats without nephrocalcinosis was younger at 6 years (range, 1-17 years; IQR, 2.75-11.5; $n = 20$; $P < .001$). Females were overrepresented among cats with nephrocalcinosis ($P = .03$; [Table 2](#)). Among the 147 cats with nephrocalcinosis, 96% (141) had bilateral nephrocalcinosis ([Table S3](#)).

Table 2.

Characteristics of cats with non-nephrocalcinosis and nephrocalcinosis.

Variables (<i>n</i> [%])		Non-nephrocalcinosis (<i>n</i> = 28)	<i>n</i>	Nephrocalcinosis (<i>n</i> = 147)	<i>n</i>	<i>P</i> -value
Age (median [IQR])		6.0 [2.75, 11.5]	20	15 [12, 17]	95	<.001
Gender (% [<i>n</i>])	Female	30% (6)	20	59% (56)	95	.03
	Male	70% (14)		41% (39)		
Breed (% [<i>n</i>])	Dlh	6% (1)	17	14.1% (13)	92	
	Dmh	18% (3)		12% (11)		
	Dsh	71% (12)		65% (60)		
	Birman			1.1% (1)		
	Ragdoll			1.1% (1)		
	Russian Blue			1.1% (1)		
	Siamese	6% (1)		3.3% (3)		
	Snowshoe			1.1% (1)		
	Tonkinese			1.1% (1)		
	None	75% (21)	28	40% (59)	147	
	Unilateral	21.4% (6)		24% (35)		
	Bilateral	4% (1)		36% (53)		
Nephrolithiasis (% [<i>n</i>])						.001

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Abbreviations: Dlh = domestic longhair; Dmh = domestic mediumhair, Dsh = domestic shorthair.

Prevalence of nephrocalcinosis in kidneys

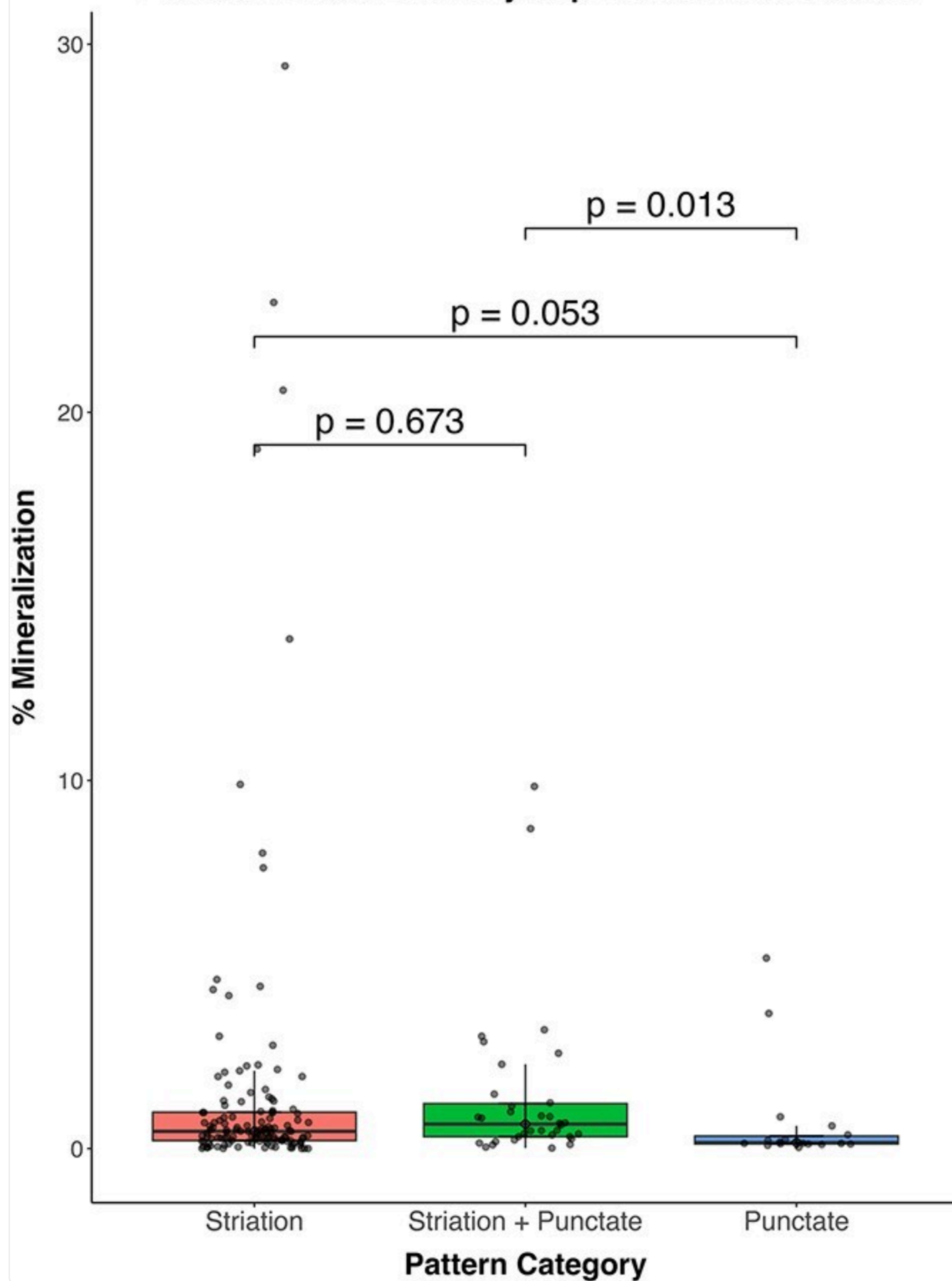
Nephrocalcinosis was observed in 82% (288/350) of kidneys. Of affected kidneys, 73% (210) showed striated nephrocalcinosis (76% [159/210] diffuse; 24% [51/210] regional); 6% (18/288) showed punctate nephrocalcinosis (83% [15/18] diffuse; 17% [3/18] regional); and 20% (57/288) had combined striated and punctate patterns (70% [40/57] diffuse striation and diffuse punctate; 23% [13/57] regional striation and diffuse punctate; and 7% [4/57] regional striation and regional punctate). In 3 kidneys (1%), nephrocalcinosis was unclassified ([Figure S1](#)).

The percentage of mineralization on microradiography correlated with micro-CT volume (Kendall's $\tau = 0.74$; $P < .001$; $n = 13$; [Figure S2](#)) and the von Kossa score (Kendall's $\tau = 0.62$; $P < .001$, $n = 32$).

Percentage mineralization was quantified in 221/350 kidneys (158 cats); 129 kidneys were not quantifiable because the central section was removed for another study before microradiography. Median percentage parenchymal mineralization was 0.5% (IQR, 0.2-1.0) in kidneys with striated nephrocalcinosis ($n = 132$), 0.2% (IQR, 0.1-0.3) in punctate ($n = 18$), and 0.7% (IQR, 0.3-1.2) in combined striated and punctate patterns ($n = 34$; [Figure 3](#)). Percentage mineralization did not differ significantly between striated and punctate patterns ($P = .05$). Combined patterns showed higher percentage mineralization than punctate ($P = .01$), but not than striated ($P = .67$). Thirty-seven kidneys had no mineralization.

Figure 3.

Percent Mineralization by Nephrocalcinosis Pattern



Box plot of percentage mineralization by nephrocalcinosis pattern. Each box represents the interquartile range with the horizontal line indicating the median. Pairwise comparisons were evaluated using Dunn's test with Bonferroni correction.

For kidneys with histopathology available ($n = 123$ kidneys from 123 cats), von Kossa regional involvement by microradiographic pattern is summarized in [Table S4](#). Microradiographic nephrocalcinosis patterns were classified as striated ($n = 74$), punctate ($n = 5$), combined striated and punctate ($n = 21$), none ($n = 22$), and unclassified ($n = 1$). Striated mineralization was associated with von Kossa-positive mineralization in medullary regions (95.8% [91/95] with striation; 82.1% [23/28] without striation; $P = .03$; OR, 4.9; 95% CI, 1.0-26.6). Punctate mineralization was not significantly associated with von Kossa-positive mineralization in cortical regions (26.9% [7/26] with punctate; 18.6% [18/97] without punctate; $P = .41$; OR, 1.6; 95% CI, 0.5-4.8).

Association between nephrocalcinosis and nephrolithiasis in cats

Among cats with nephrocalcinosis, 60% (88/147) had nephroliths, compared with 25% (7/28) among cats without nephrocalcinosis ($P = .001$). In the subset with quantitative microradiography ($n = 158$), percentage mineralization was higher in cats with nephroliths (median, 0.5%; IQR, 0.2%-0.9%; $n = 75$) than in those without (median, 0.3%; IQR, 0%-0.6%; $n = 83$; $P = .004$).

In univariate logistic regression analysis, nephrocalcinosis was associated with nephrolithiasis (OR, 3.5; 95% CI, 1.6-7.4; $P = .002$; $n = 175$; [Table 3](#)). Bilateral nephrocalcinosis showed a stronger association (OR, 5.1; 95% CI, 2.0-12.8; $P < .001$), whereas unilateral nephrocalcinosis was not significantly associated (OR, 2.3; 95% CI, 0.4-12.6; $P = .36$). Age was associated with nephrolithiasis (OR, 1.2; 95% CI, 1.1-1.3; $P = .001$; $n = 115$).

Table 3.

Logistic regression analysis examining variables associated with nephrolithiasis in cats.

Variables	Univariate analysis				Multivariable analysis			
	OR	95% CI	P-value	n	OR	95% CI	P-value	n
Age	1.2	1.1-1.3	.001	115	1.1	1.0-1.2	.04	115
Sex	1.4	0.3-7.0	.67					
Nephrocalcinosis	3.5	1.6-7.4	.002	175	3.1	1.0-10.1	.05	
Unilateral nephrocalcinosis	2.3	0.4-12.6	.36					
Bilateral nephrocalcinosis	5.1	2.0-12.8	<.001					

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In multivariable logistic regression adjusting for age, kidneys with nephrocalcinosis had an OR of 3.1 for nephrolithiasis (95% CI, 1.0-10.1; $P = .05$; $n = 115$). Age remained associated with nephrolithiasis (OR per year, 1.1; 95% CI, 1.0-1.2; $P = .04$; [Table 3](#)).

Association between nephrocalcinosis and nephrolithiasis in individual kidneys

Among kidneys with nephrocalcinosis, 49% (141/288) had nephroliths compared with 13% (8/62) of kidneys without nephrocalcinosis ($P < .001$). Nephrolith prevalence did not differ among striated, punctate, and combined patterns ($P = 1.00$). However, the prevalence of nephrolithiasis for each pattern was higher compared to kidneys without nephrocalcinosis ($P < .001$). In 221 kidneys, percentage mineralization was higher in kidneys with nephroliths (median, 0.4%; IQR, 0.2%-1.0%; $n = 101$) than in those without (median, 0.2%; IQR, 0%-0.7%; $n = 120$; $P = .01$).

In cases where age was provided, the association between nephrocalcinosis patterns and nephroliths was evaluated using 2 mixed effects logistic regression models ([Figure S3](#)). Age was included as a fixed effect, and the individual cat was included as a random effect to account for within-cat clustering. Nephrocalcinosis was present in 186 of 230 kidneys.

In the first model, kidneys were categorized into 3 mutually exclusive groups: striated, punctate, and combined striated and punctate. Striated nephrocalcinosis (OR, 13.9; 95% CI, 2.0-95.9; $P = .01$) and the combined pattern (OR, 18.6; 95% CI, 2.2-160; $P = .01$) were associated with higher odds of nephrolithiasis. For punctate nephrocalcinosis, the estimate was imprecise and did not reach significance (OR, 6.7; 95% CI, 0.2-26; $P = .30$). Age was not significantly associated with nephrolithiasis in this model (OR, 1.1; 95% CI, 1.0-1.3; $P = .11$).

In the second model, kidneys were grouped by the presence of striated or punctate nephrocalcinosis regardless of overlap; groups therefore were not mutually exclusive. Striated nephrocalcinosis remained associated with nephrolithiasis (OR, 10.0; 95% CI, 1.8-54.8; $P = .01$). In contrast, punctate nephrocalcinosis was not significantly associated with nephrolithiasis (OR, 1.7; 95% CI, 0.4-6.7; $P = .44$). Age was not significant in this model (OR, 1.1; 95% CI, 1.0-1.3; $P = .07$).

Discussion

Nephrocalcinosis co-occurred frequently with nephrolithiasis in cats. Among kidneys with nephrocalcinosis, 49% had nephrolithiasis. The biological mechanisms linking these 2 conditions remain incompletely understood, but may arise from shared underlying causes. These 2 forms of pathological mineralization are commonly observed in CKD, suggesting that systemic factors such as renal injury, dysregulated mineral metabolism and excretion, or advancing age may promote development of both conditions.^{[3,4](#)} Nephrocalcinosis may represent a plausible catalyst facilitating nephrolith formation. However, our study cannot establish if nephrocalcinosis preceded nephrolithiasis.

In humans, a widely accepted hypothesis proposes that CaP deposits within the renal interstitium (Randall's plaque) or renal tubules (Randall's plug), once exposed to urine, serve as a nidus for CaOx stone formation.^{[10,12-14](#)} In cats, striated nephrocalcinosis appears morphologically consistent with this hypothesis, and results of our study indicated that a striated pattern was associated with the presence of nephrolithiasis. If linear tracks of parenchymal mineralization become exposed to the urinary space, they too can serve as a site for initial nephrolith formation. In support of this hypothesis, some CaOx nephroliths in cats contain CaP cores resembling calcified tubules as confirmed by one nephrolith in this study ([Figure 2](#)).^{[15-17](#)} Consistent with this

concept, microradiographic striation was associated with von Kossa–positive mineralization in the papilla or mid-medulla in the histopathology subset, supporting a predominantly medullopapillary distribution. In contrast, punctate nephrocalcinosis, which likely reflects dystrophic calcification secondary to vascular or glomerular injury, was not associated with nephrolith formation.¹⁸ The microradiographic punctate pattern was not significantly associated with von Kossa staining in the cortical or corticomedullary junction areas, likely reflecting limited statistical power given the small size of the punctate-only group. Although causality cannot be established, the predominance of medullopapillary mineralization and its anatomic proximity to the collecting system emphasize the papilla as a plausible site where tissue mineralization may provide a substrate for nephrolith anchoring and growth.

Alternatively, nephrocalcinosis and nephrolithiasis may also arise from distinct but overlapping mechanisms. Both are common in older cats, suggesting that they may represent parallel outcomes of age-related renal injury.^{3–7} Nephrocalcinosis typically involves calcium deposition within the renal interstitium or renal tubules, whereas nephrolithiasis depends on crystal retention within the urinary space. A kidney may exhibit diffuse mineralization without developing a nephrolith if other contributing factors such as urinary supersaturation or altered flow dynamics are absent.

The 54% prevalence of nephrolithiasis identified in our study aligns with previous reports.^{4,19} Although this percentage is higher than the 7% prevalence reported in a study of 693 hospitalized cats in the United States, the discrepancy likely reflects differences in study populations and diagnostic sensitivity.²⁰ Our cohort was older (median, 14 years) than the general hospitalized population (median, 10 years), and we utilized high-resolution microradiography. This advanced imaging technique allowed for the detection of subtle or subclinical mineralization that often is missed by standard clinical radiography and ultrasonography. These findings emphasize that nephrolithiasis is a substantial burden in the aging feline population, particularly considering uroliths were identified as the cause of obstruction in 73% of affected kidneys.²¹

Calcium oxalate is the most common nephrolith type in cats. From 2022 to 2023, the Minnesota Urolith Center analyzed uroliths from over 45,000 cats; 765 submissions were from the kidney or ureter. Seventy-six percent of upper tract uroliths were composed of CaOx.²² A historical analysis of 2445 upper urinary tract uroliths similarly reported that 70% were CaOx.^{23,24} These findings emphasize the clinical importance of CaOx nephrolithiasis. In our study, urolith composition was available for a small randomly selected subset ($n = 10$), and 5/10 nephroliths were calcium phosphate. Given the limited sample size, these results are not intended to estimate composition prevalence in the broader cohort.

Our retrospective study had some limitations. The kidneys examined were obtained postmortem from cats that were euthanized, often because of chronic or terminal illness. As such, our findings may overrepresent the prevalence of renal mineralization compared to healthy or younger cats. Nonetheless, our study provided valuable access to a large number of intact kidneys, enabling detailed evaluation of mineral patterns and burden across disease stages. Although high-resolution microradiography enabled sensitive detection of nephroliths and nephrocalcinosis, it might not detect non-radiopaque uroliths. In addition, A small central section was removed from some kidneys before imaging. Overall mineralization patterns remained assessable, but focal mineralization or small calculi confined to the excised region may have been missed. Nephrocalcinosis refers to the deposition of calcium salts, typically CaP or CaOx, within renal tissue.²⁵ In our study, von Kossa staining confirmed the presence of calcium salts. Previous studies using both von Kossa and Alizarin Red S staining have indicated that nephrocalcinosis in cats is primarily composed of CaP.²⁶ Clinicopathologic data were not available at the time of euthanasia limiting our ability to use biochemical data for assessing mineral balance. Estimates for punctate nephrocalcinosis are imprecise because of the small number of kidneys with punctate nephrocalcinosis. The cross-sectional design precludes temporal inference.

In conclusion, nephrocalcinosis was associated with nephrolithiasis in cats. Striated nephrocalcinosis patterns were associated with nephroliths in individual kidneys. Future studies may help clarify the mechanistic relationship between striated mineralization and nephrolith formation. Understanding the mechanisms underlying striated mineralization and mitigating its development may offer a potential target for the prevention of nephrolithiasis in cats.

Supplementary Material

aalag092_Supplemental_Files

[aalag092_supplemental_files.zip](#) (1MB, zip)

Abbreviations

3D

three-dimensional

ANOVA

analysis of variance

CaOx

calcium oxalate

CaP

calcium phosphate

CKD

chronic kidney disease

CT

computed tomography

Dlh

domestic long-hair

Dmh

domestic medium-hair

Dsh

domestic short-hair

micro-CT

micro-computed tomography

OR

odds ratio

P

P-value

PPI

pixels per inch

ROI

region of interest

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Nuttha Hengtrakul (Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Writing—original draft, Writing—review & editing), Eva Furrow (Supervision, Validation, Writing—review & editing), Ferenc Toth (Supervision, Validation, Writing—review & editing), Michael Borofsky (Supervision, Validation, Writing—review & editing), and Jody P. Lulich (Conceptualization, Funding acquisition, Methodology, Supervision, Validation, Writing—review & editing)

Conflicts of interest

The authors declare no conflict of interest.

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Off-label antimicrobial declaration

The authors declare no off-label use of antimicrobials.

Institutional animal care and use committee or other approval declaration

This study utilized organs from deceased animals and is exempt from institutional animal care and use committee review. All owners provided written informed consent for their animal's remains to be used for research purposes.

Human ethics approval declaration

The authors declare that human ethics approval was not needed.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

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